

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E. (E&E) (Credit System) Degree Examinations**

May- June 2023

19EEE235 – ELECTRICAL ESTIMATION AND COSTING

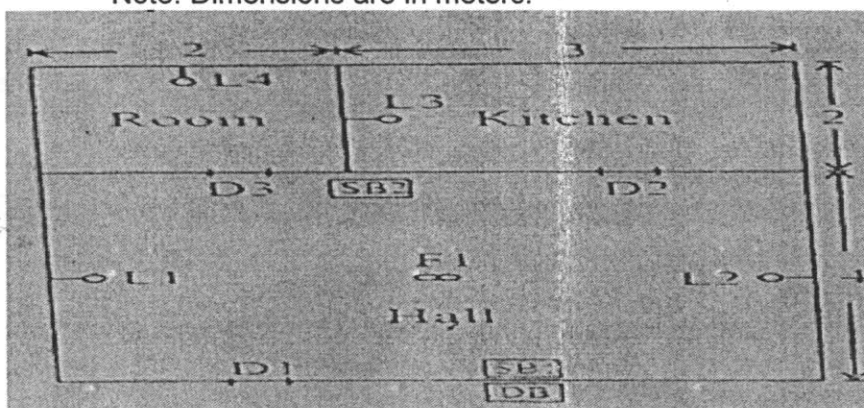
Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a) What are the factors to be considered for the choice of wiring system? 06 L*1 1 1
- b) List the data required for preparing the electrical estimate of interior wiring. 06 L2 2 1
- c) List the general rules for interior wiring. 08 L3 1 2
2. a) Explain in detail the steps to be carried out for estimation of interior wiring 08 L2 2 2
- b) Small flat plan is as shown below. It is required to provide the electrical connections. The fittings are fixed as shown in plan.
- Calculate number of sub-circuits.
 - Calculate size and length of the wire.
 - Estimate the quantity of required materials. Adopt PVC conduit wiring.
- Note: Dimensions are in meters.

12 L4 3 3

3. a) Explain the following Indian Electricity (IE) Act and IE rules as required by the general safety requirements:
- IE Rule29
 - IE Rule30
 - IE Rule45
 - IE Rule46
- 10 L3 1 1
- b) Draw the neat diagram of plate earthing with all dimensions as per IE rules. Also mention the application of pipe and plate earthing. 10 L3 1 3

P.T.O.

Unit – II

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| 4. a) Mention the concept of Service Connection.
List the specifications of materials used in UG cable service connection.
Also state & explain the different methods of laying the UG Cable. | 10 | L2 | 3 | 3 |
| b) List and mention the specifications of materials used in 11kV Feeder | 10 | L3 | 2 | 2 |
| 5. a) Mention the important considerations required for motor installation wiring | 08 | L2 | 2 | 3 |
| b) Squirrel cage Induction motor 20HP, 3 phase, 50Hz, 400V, 1450 RPM is to be installed in a factory at a distance of 20 Meters from the available mains list the quantity of materials required. Surface conduit wiring is to be adopted. | 12 | L4 | 3 | 3 |
| 6. a) Draw the neat diagram & clearly mentioning the items/parts for following type of service connection:
i. Over head service connection &
ii. Underground service connection | 10 | L3 | 3 | 3 |
| b) Write the technical note for the following.
i. Clearance of conductors required for over head lines (both low and high voltage) from ground as per the recommendation of IE rule 77 and 79.
ii. Spacing between the conductors (Both horizontal and vertical spacing, and between conductors and supporting structures) | 10 | L3 | 4 | 3 |

Unit – III

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|--|----|----|---|---|
| 7. a) Draw the neat key diagram of typical 33 kV substations mentioning the key points. | 10 | L2 | 4 | 5 |
| b) Obtain the graphical symbols for various types of apparatus and circuit elements required for substations electrical connection.
Also write a note on main electrical connections of a substation. | 10 | L2 | 4 | 5 |
| 8. a) Write a note on the following:
i. Explain substation auxiliary supply with the neat key diagram.
ii. Substation earthing with dimensional figure and explanation | 10 | L2 | 5 | 5 |
| b) List and explain the equipments required for substations. | 10 | L3 | 3 | 5 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
